Control Systems – Key Terms

Embedded Control of Robotics – revision notes on transfer function analysis, time response and Laplace basics

1. Poles and zeros

- **Zeros:** values of s that make the numerator N(s) = 0 (the transfer function becomes zero).
- **Poles:** values of s that make the denominator D(s) = 0 (the transfer function becomes infinite).
- Poles decide stability and the form of the transient response; zeros shape the response.

Finding them

```
G(s) = (s + 1) / ((s + 2)(s + 3))
Zero: s + 1 = 0 → s = −1
Poles: s = −2, s = −3   → all negative → stable
```

Plotting on the s-plane

The horizontal axis is Re(s) and the vertical axis is Im(s). Poles are marked × and zeros ○.

- Real poles or zeros (no j term) lie on the real axis. Example: a pole at −2.
- Complex values lie off the axis as a conjugate pair. Poles at −2 ± j3 are plotted at (−2, 3) and (−2, −3).
- A zero at 2 + j sits at (2, 1). Nothing is plotted on the imaginary axis unless the real part is zero.
- Values such as ±j3 (real part 0) lie on the imaginary axis.

Pole location	Response
Left half plane (Re < 0)	Dies out – stable
Imaginary axis (Re = 0)	Keeps oscillating – marginally stable
Right half plane (Re > 0)	Grows without limit – unstable

The real part of a pole sets how quickly the response decays or grows; the imaginary part sets the oscillation frequency.

Effect of zeros

Zeros do not make a system unstable, because stability depends only on the poles. A zero in the right half plane (for example s = +1 from a factor (s − 1)) gives a non-minimum-phase response: after a step input the output first moves the wrong way, then settles. Example: for (s − 1)/((s + 2)(s² + 4)) the poles are −2 and ±j2, so the system is marginally stable whatever the zero does.

Gain

A constant gain K multiplies the whole transfer function and is written in the numerator. "A gain of 2 with a zero at −1 and poles at 0 and −2" gives G(s) = 2(s + 1) / (s(s + 2)). The gain scales the output but does not move the poles or zeros.

2. Time-domain analysis

- **Time domain:** signals described as functions of time t (output vs time).
- **Frequency domain:** signals described by frequency ω, i.e. how the system responds to slow and fast inputs.
- **Time response:** how the output changes over time after an input such as a step. It has a **transient** part, which dies out and contains any overshoot or oscillation, and a **steady-state** part, the final settled value.
- **DC gain:** the steady-state output for a constant (zero-frequency) input, G(0). An input of 1 V giving 2 V means a DC gain of 2.
- **Time constant τ:** time to reach 63% of the final value in a first-order system.
- **Phase shift:** how far the output lags or leads the input. **Amplitude:** the size of the output.

Step-response specifications

Term	Meaning
Delay time t_d	Time to reach 50% of the final value
Rise time t_r	Time to go from 10% to 90% of the final value
Maximum overshoot M_p	Amount by which the output exceeds its final value, usually in percent
Peak time t_p	Time at which the maximum overshoot occurs
Settling time t_s	Time after which the output stays within 2% (or 5%) of its final value
Steady-state error e_ss	Difference between the desired and the final output

Damping ratio ζ (zeta)

ζ	Case	Response
0	Undamped	Oscillates forever
0 < ζ < 1	Underdamped	Oscillates, then settles
1	Critically damped	Fastest settling without oscillation
> 1	Overdamped	Slow, no oscillation

3. Frequency analysis

Replace s by jω to get the frequency response G(jω). Its magnitude |G(jω)| and phase ∠G(jω) show the gain and phase shift at each frequency, the resonance frequency, and

the gain and phase margins used to judge stability (Bode plots). ω_n is the natural frequency.

4. Feedback

```
Negative feedback:  C/R = G / (1 + GH)       Unity:  C/R = G / (1 + G)
```

The feedback is negative when the feedback signal enters the summing junction with a minus sign.

5. Laplace transform rules

```
y(t) → Y(s)
dy/dt → sY(s)            (zero initial conditions)
d²y/dt² → s²Y(s)
∫ y dt → Y(s)/s
```

Differentiation multiplies by s and integration divides by s.

Circuit impedances in the s-domain

```
Resistor  R      Inductor  sL      Capacitor  1/(sC)
```

For a capacitor $v = (1/C)\int i\,dt$, so $V(s) = I(s)/(sC)$, and $1/(sC)$ acts as its impedance. For a circuit, the transfer function is the output voltage over the input voltage, $V_o(s)/V_i(s)$.

Reading a first-order step response

If the output rises smoothly from 0 to a final value of 2 without overshoot, the system is first order: $y(t) = K(1 - e^{(-t/\tau)})$ with $K = 2$. Read τ as the time at which $y = 0.632 \cdot K$, then $G(s) = K/(\tau s + 1)$.

6. Effect of adding poles and PID

- Adding poles makes the system slower and less stable; adding a left-half-plane zero tends to speed it up.
- P reduces rise time, I removes steady-state error, D reduces overshoot. $C(s) = K_p + K_i/s + K_d \cdot s$.
- LHP = left half plane (stable region); RHP = right half plane (unstable region).

7. Root locus

The root locus shows how the closed-loop poles move on the s-plane as the gain K increases from 0 to ∞. Branches start at the open-loop poles and end at the open-loop zeros (or go to infinity). While every branch stays in the left half plane the closed-loop system is stable; the gain at which a branch crosses the imaginary axis is the stability limit.